

Poorvi Hosakere Chandrashekhar

Github, Personal Website, LinkedIn

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Education	International Institute of Information Technology - Hyderabad <i>Dual Degree Program:</i> B.Tech in CS and MS by Research in CNS Master of Science , Computational Natural Sciences Bachelor of Technology with Honors , Computer Science and Engineering	Nov 2021 – May 2026 Overall GPA: 8.8/10.0
Scholastic Achievements	Google Women Engineers (WE) Scholar: Selected as a Top 100 Scholar nationwide for a 2-year technical mentorship program by Google; awarded a scholarship of INR 100,000. Uber She++ (UberSTAR): Selected as one of the top 35 women engineers from Tier 1 engineering schools in India for the coveted UberSTAR internship program. KVPY Fellow (All India Rank 298): Awarded the prestigious Kishor Vaigyanik Protsahan Yojana fellowship by the Dept. of Science and Technology, Govt. of India (Administered by IISc Bangalore) for aspiring researchers. Megathon Winner: Secured 1st place (Team of 5) in Hyderabad's largest student-run hackathon (Prize: INR 30,000). Indian Olympiad Qualifier in Biology (IOQB): Achieved state-level distinction as one of the Top 13 students in Karnataka selected for the prelims.	
Publications	HC Poorvi , PK Vinod. (2025) Integrated Cross-Organ Transcriptomic Analysis Uncovers Conserved Gene Signatures Predictive of Allograft Rejection. <i>Under Review at BMC Bioinformatics Journal</i> . bioRxiv Gaurav Bhole*, HC Poorvi* , JR Madhav*, Prabhakar Bhimalapuram, PK Vinod. (2025) HyperCLSA: Hypergraph Contrastive Learning with Self-Attention for Multi-Omics Integration. Accepted at <i>11th International Conference on Pattern Recognition and Machine Intelligence</i> . [Oral Presentation] bioRxiv	
Research Experience	Research Assistant May 2024 – Present Lead Researcher & First Author: Integrated Cross-Organ Transcriptomic Analysis (Manuscript under review at <i>BMC Bioinformatics</i>). - Conceptualized a unified molecular framework for allograft rejection by integrating transcriptomic data from 672 biopsies across liver, kidney, and heart, challenging traditional organ-specific diagnostic paradigms. - Constructed organ-specific gene co-expression networks using Leiden clustering and Jaccard similarity to map conserved regulatory modules, identifying a consensus six-module immune cascade (172 genes) capturing the alloimmune response hierarchy. - Uncovered a novel, highly conserved 24-gene cell cycle signature upregulated in rejecting allografts, establishing cellular proliferation as a core, previously underappreciated mechanism of rejection pathology. - Engineered a machine learning pipeline with SHAP consensus feature ranking to distill high-dimensional data into a robust 20-gene universal biomarker panel ($AUC > 0.96$) - Validated signatures on an independent lung transplant cohort ($n = 243$, $AUC > 0.97$), benchmarking against established RATs panels.	Centre for Computational Natural Sciences and Bioinformatics (CCNSB) Lab, IIIT-H Advisor: Dr. Vinod PK
	Research Assistant Jan 2025 – July 2025	CCNSB Lab, IIIT-H Advisors: Dr. Vinod PK & Dr. Prabhakar Bhimalapuram
	Co-First Author: HyperCLSA (Accepted at <i>PReMI 2025</i> for Oral Presentation ; to appear in Springer LNCS Series).	

- Co-developed **HyperCLSA**, a novel deep learning framework for multi-omics breast cancer subtyping, integrating **hypergraph convolutional networks (HGNN)** with **supervised contrastive learning** to align heterogeneous latent spaces.
- Engineered a robust pipeline using **Boruta feature selection** and **radius-based hypergraph construction** to capture high-order gene correlations, followed by **multi-head self-attention** for adaptive modality fusion.
- Achieved state-of-the-art **90.1% accuracy** on the **TCGA-BRCA dataset** ($n = 850$), significantly outperforming established graph-based baselines (MOGONET, MORE) in PAM50 subtype classification.

Research Assistant

July 2025 – Present

Cognitive Science Lab, IIIT-H

Advisor: Dr. Bapi Raju

Project: Investigating Structural-Functional Coupling in Task-Based fMRI via Graph Diffusion Wavelets.

- Investigating the generalizability of the **Graph Diffusion Wavelet** framework (originally developed for resting-state SCFC mapping) to task-specific fMRI contexts.
- Testing the hypothesis that **ROI-specific diffusion scales** – which govern information flow on the structural connectome—adapt dynamically under varying cognitive loads (e.g., language processing, motor tasks).
- Benchmarking model performance on the **Human Connectome Project (HCP)** dataset to analyze deviations in linear structure-function coupling across distinct brain states.

Industry Experience **Software Engineering Intern**, Bengaluru, India

May 2025 – July 2025

Stripe

Ruby, Typescript, MongoDB, Jest

Developed an end-to-end, user-facing multi-currency charge lookup flow within the Support Chatbot to automate transaction inquiries.

- Productionized the feature to 100% of global users, reducing support case creation by **~120/week** by enabling self-resolution of unrecognized transactions.
- Utilized state-driven workflows to securely handle **sensitive financial data**, ensuring compliance with enterprise security standards.
- Ensured system reliability by implementing comprehensive unit tests (~90% coverage) and authoring technical documentation to bridge knowledge gaps.
- **Received Return Offer** for a full-time Software Engineer role based on exceptional performance and successful feature deployment.

Software Engineering Intern

May 2024 – July 2024

Uber

GO, Typescript, Redux, Fusion.js, Jest

Engineered a full-stack, agent-facing central dashboard using Fusion.js (React) and Go to streamline global entity verification processes.

- Architected a **configuration-driven UI framework** where JSON definitions dynamically render views, eliminating the need for hardcoded region-specific logic.
- **Drastically reduced engineering overhead** for new market launches by abstracting complex regulatory requirements into simple boolean flags, enabling rapid scaling across user, role and environment types.
- **Received Return Intern Offer** based on successful deployment of the dashboard and technical excellence.

Teaching Experience **Head Teaching Assistant, Computer Programming (CSO.101)**: Monsoon 2025.

Instructors: Prof. Abhishek Deshpande, Prof. Vineet Gandhi, Prof. Girish Varma.

Teaching Assistant, Statistical Methods in AI (CS7.403): Spring 2025.

Instructor: Prof. Vineet Gandhi.

Teaching Assistant, Computer Programming (CSO.101): Monsoon 2024.

Instructors: Prof. Girish Varma, Prof. Abhishek Deshpande.

Teaching Assistant, Intro to Software Systems (CS6.201): Spring 2024.

Instructors: Prof. Anoop Namboodiri, Prof. Prakash Yalla.

Teaching Assistant, Algorithm Analysis and Design (CS1.301): Monsoon 2023.
Instructor: Prof. Suryajith Chillara.

Course Projects

CS4.409 Data Foundation Systems Project Aug 2023 – Dec 2023
Indian Driving Dataset (IDD) Filtering Engine iHub-Data (DST, Govt. of India)
Engineered a semantic search and filtering platform using the MERN stack, enabling researchers to query massive Indian Driving Datasets (IDD) based on granular metadata (object classes, image dimensions, scene tags).

- **Architected high-performance retrieval pipelines** by integrating **Elasticsearch** for sub-second metadata querying and **MinIO** for scalable object storage, allowing users to generate and download custom dataset subsets.
- Collaborated with iHub-Data stakeholders to refine requirements, demonstrated the tool to multiple research labs, and successfully deployed the system on the official Data Foundation portal.

CS3.301 Operating Systems and Networks Aug 2023 – Nov 2023
Project
Distributed Network File System & C-Shell
Built core system infrastructure from scratch using C and TCP/IP socket programming to replicate fundamental OS functionalities.

- **Network File System (NFS):** Architected a resilient distributed file system implementing a central Naming Server and concurrent Storage Servers to streamline file operations across a distributed network.
- **Custom C-Shell:** Designed a modular Unix shell supporting advanced command parsing, I/O redirection, piping, and signal handling, replicating core Bash functionality.
- **Optimization:** Executed concurrent client access handling and optimized search mechanisms using semaphores, improving overall system throughput.

Leadership

Women in Tech Club (LeanIn), *Coordinator* Aug 2024 – Sep 2025
Entrepreneurship Cell (E-Cell), *Vice Coordinator* Aug 2023 – Aug 2024
Entrepreneurship Cell (E-Cell), *Content Head* Aug 2022 – Aug 2024
IIIT-H College Fest (Felicity), *Content Team Lead* Jan 2024 – Apr 2024
Counseling Cell, *Apex Mentor (Freshmen Coordinator)* Aug 2022 – Aug 2023
Christel House India NGO (Nokia Partnership Project), *Volunteer Tutor* Jan 2022 - Aug 2022

Skills

Languages: C, Python, R, C++, Ruby, Go, SQL, Bash, Assembly, L^AT_EX.
Web Development: HTML, CSS, JavaScript, TypeScript, React.js, Redux, Node.js, Express.js, Fusion.js, PHP, Bootstrap, MERN Stack.
Applications & Databases: Git/GitHub, Vi/Vim, VS Code, MongoDB Atlas, Elasticsearch, MinIO.
Operating Systems: Linux (Ubuntu), macOS, Windows, Android.